

process technologies and physics of deep-submicron devices. Given that the Indian semiconductor industry is mostly design oriented, with minimal experience in IC fabrication, will the EDA industry be curtailed in its growth? Can Indian universities step up to play a significant role to groom the EDA industry, similar to the roles played by Stanford and UC Berkeley during the EDA industry's infancy?

It is clear that the absence of EDA products 'made in India' from the large EDA companies (Cadence, Synopsys, Mentor and Magma) was perceived by many to be a disincentive to the graduates in this field. This is, however, not entirely true, as there are EDA products made in India. For example, much of Atrenta's R&D occurs in India, leading to some of its products being designed in India. A major new EDA product has also been developed by Mentor in India, with its key customer being the Broadcom design centre in India. Perhaps, the EDA companies need to do a little more publicity about their products, to boost the morale and recruitment in the Indian arena.

Moving to new edge

In a couple of years, lots of designs will shift from the present 90- and 65-nanometre process to 45 nanometres. The transition to 45nm has already begun. "Indian design service companies are gaining an increasing

Evolution of design life-cycle

The life cycle depends on the design. If I have to design a microprocessor for Intel or some big company, the design time is typically years. If I start designing today a microprocessor, it will take two years before I will be able to bring it to the market and then whether the market accepts it or not is a different issue. So it takes a lot of investment, whether you design a microprocessor or a DRAM.

Typically, we do not have that much time, if we have to design a new multimedia recorder or iPhone, to design all the chips from scratch. In the 1990's we still did customise design or cell-based design or ASIC design. Nowadays, ASIC design is also very complicated as it takes six months to one year, which again is very slow. If I have to make a competitor of iPhone, I will have to get the money and design the chip around. Then I need to complete the whole work within 10-12 months. So for designing the chip, I have less than 3-6 months. So when you have that very little time, what you do for a digital design is only FPGA-based designing. At the most, you design an ASIC. As soon as you prove the design on FPGA, you prototype it and can translate it into ASIC with some modifications using software.

Ten years ago, if I were to design a chip, the actual architecture of it was very important. So people used to spend a lot of time and money on designing the chip and it was very difficult for someone to compete with them because of the intensive design part. But now time is very critical, so if you have to design the next-generation cell phone you have to come up very quickly within six months. So people prefer not to spend so much of time in design process.

Once I have designed it and worked on it, I have to test the chip using simulation tools. Basically, after designing we have EDA tools. Earlier design was done in logic level or transistor level but now everything is done in RTL level.

The main difference between doing chip design ten years ago and now is the time factor. People do not care if the design is not 100 per cent efficient or the best. They are just interested in that the design works and that they finish it within the number of months that they have decided. The competition is so high that if you do not come to the market in ten months, somebody else will. The design may not be efficient but as long as it works it is fine.

If Intel were to design a microcontroller and if they were to make it 10 per cent more efficient in terms of power and faster, they could charge much more money for the microprocessor and they knew the CPU chip was going to be running for several years. So over the life of the chip, that 10 per cent more power efficiency or 10 per cent faster performance of the chip was very important and very critical. Whereas, if I have a wireless router that is 10 per cent less efficient, you will not even notice it.

—As told by Manish Sharma, associate professor, Centre for Applied Research in Electronics, IIT Delhi, to EFY's Abhijit Paul Chaudhury

share of designs in advanced process nodes. Designs in new process nodes bring their own set of challenges with them, but at the same time, as they are

cutting-edge, they also result in fatter margins for vendors," says Roy.

The semiconductor integrated circuits (IC) industry is driven by the need for small, low-cost, high-performance devices. The impact of Moore's Law as well as the increasing demand from the consumer electronics and telecom sectors are all fueling the need for miniaturisation and the move towards ever-shrinking form factors. Different types of materials and device configurations are being investigated to reduce the size of a chip without compromising on the functionalities of the device.

For example, scaling is an issue with eDRAMs (embedded dynamic random access memories) below the 45nm node. However, by using a different configuration of eDRAM, such as capacitorless

"Taking inspiration from the Nano, if we start building more products they are bound to be successful anywhere in the world. This is because the products designed in India, for India, are made keeping many constraints in mind while also considering the high demand."

—Jayaram Pillai, managing director for India, Russia & Arabia, National Instruments

